Dipesh Biswa

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EDUCATION

Rochester Institute of Technology, Golisano College of Computing and Information **Rochester, NY**

Bachelor of Science in Software Engineering *Anticipated May 2029*

- Recipient of the RIT Presidential and Davitt Endowed Scholarship.

Relevant Coursework

• Engineering of Software Subsystem • Engineering of Enterprise SW System • Human Centered Requirements and Design • Software Process and Project Management

PROJECT EXPERIENCE

Echo - Offline AI-Powered Tourism Feedback App. *October 2026*

7th Hack-Nation Global AI Hackathon - New York Hub (Columbia)

- Co-developed backend and AI workflows for a tourism feedback application that works offline, helping small operators identify visitor preferences and improve their services.
- Implemented offline translation from English, German, French, and Italian into Kiswahili to make multilingual visitor feedback accessible to local operators.
- Co-developed topic summaries, evidence-backed recommendations, and improvement tracking, with uncertain AI predictions flagged for human review.

IoT Houseplant Health Monitor (Personal) *Aug 2026- Sep 2026*

- Collaborated with an embedded hardware engineer to define RESTful API contracts and JSON payload specifications, enabling real-time telemetry ingestion from an ESP32 microcontroller
- Architected a Spring Boot backend with Spring Data JPA and PostgreSQL to validate, deserialize, and persist incoming environmental metrics (soil moisture, sunlight, temperature, humidity)
- Integrated Anthropic Claude API to construct dynamic prompts from stored telemetry data, generating automated plant health diagnostics and actionable care recommendations
- Deployed entire project on Railway to receive and persist telemetry from an ESP32 over HTTP

Personal Finance Tracker (Personal) *Dec 2025- Jan 2026*

- Built a full-stack finance application with Spring Boot and React, integrating the Plaid API to synchronize and categorize banking transactions.
- Implemented JWT authentication with Spring Security to restrict access to protected REST endpoints and user financial data.
- Automated transaction retrieval and processing from Plaid, storing banking records in PostgreSQL for tracking within the application.

EXPERIENCE

YOH (Contractor at L3Harris Technologies) **Rochester, NY**

AOI Operator *April 2025- August 2025*

- Operated and optimized AOI inspection systems to detect defects in PCB assemblies
- Collaborated with engineers and technicians to resolve inspection discrepancies, strengthening problem-solving, teamwork, and technical communication skills.
- Documented inspection results and maintained records in company databases, enhancing accuracy, attention to detail, and regulatory compliance support.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, HTML/CSS, SQL
Frameworks & Libraries: Spring Boot, Spring Security, Spring Data JPA, React, JUnit, Jest
Databases: PostgreSQL, MySQL, MongoDB
Tools & Technologies: Git, REST APIs, GitLab CI/CD, Maven, Railway, Render